
A Multi-Task Benchmark for Abusive Language Detection in Low-Resource Settings

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Abstract

Content moderation research has recently made significant advances, but still fails to serve the majority of the world’s languages due to the lack of resources, leaving millions of vulnerable users to online hostility. This work presents a large-scale human-annotated multi-task benchmark dataset for abusive language detection in Tigrinya social media with joint annotations for three tasks: abusiveness, sentiment, and topic classification. The dataset comprises 13,717 YouTube comments annotated by nine native speakers, collected from 7,373 videos with a total of over 1.2 billion views across 51 channels. We developed an iterative term clustering approach for effective data selection. Recognizing that around 64% of Tigrinya social media content uses Romanized transliterations rather than native Ge’ez script, our dataset accommodates both writing systems to reflect actual language use. We establish strong baselines across the tasks in the benchmark, while leaving significant challenges for future contributions. Our experiments reveal that small, specialized multi-task models outperform the current frontier models in the low-resource setting, achieving up to 86% accuracy (+7 points) in abusiveness detection. We make the resources publicly available to promote research on online safety.¹

1 Introduction

The proliferation of social media has revolutionized global communication, enabling unprecedented connectivity while simultaneously creating new vectors for harm through abusive content [47]. Online hostility and harassment affect millions of users, particularly vulnerable groups including minors and minority communities, often causing physical and psychological harm while reinforcing social marginalization [14, 17]. Although significant progress has been made in automated detection of abusive content for high-resource languages such as English [11, 61, 30], the majority of the world’s low-resourced languages such as those spoken in Africa remain severely under-served [38], creating an alarming disparity in online safety and protection.

Tigrinya, a language with approximately 10 million speakers mainly in Eritrea and Ethiopia, exemplifies this technological divide [15]. Despite its significant speaker population, Tigrinya remains computationally under-resourced with minimal datasets, tools, and models [21], and in particular, there is a lack of well-established benchmarks to gauge progress in content moderation research. This gap exposes the Tigrinya-speaking communities to unchecked online abuse, and more broadly highlights the urgent need for dedicated computational resources and methods of content moderation research in underrepresented languages.²

¹ TiALD Dataset and Code: <https://github.com/fgaim/tigrinya-abusive-language-detection>

² Monitoring and Analyzing Hate Speech is a major commitment of the United Nations’ Action Plan: https://www.un.org/en/genocideprevention/documents/advising-and-mobilizing/Action_plan_on_hate_speech_EN.pdf, accessed on 2025-04-19.

In this paper, we address the critical gap by introducing the **Tigrinya Abusive Language Detection (TiALD)** dataset, a large-scale human-annotated multi-task benchmark for abusive language detection in Tigrinya. TiALD adopts a multifaceted approach, providing joint annotations for three tasks: abusiveness detection, sentiment analysis, and topic classification across 13,717 manually annotated YouTube comments. We further enrich the dataset by generating descriptions of the visual content of the corresponding videos using a Vision-Language Model (VLM), enabling the relational analysis against the user comments. Our annotation scheme enables richer contextual understanding of abusive content, supporting more nuanced analysis than that with the typical binary classification setups alone. The contributions of this work can be summarized as follows:

- We present the first large-scale multi-task benchmark dataset for abusive language detection in Tigrinya based on user comments collected from YouTube channels in the community.
- We propose a data selection methodology for iterative semantic clustering of terms to address the inherent imbalance in abusive vs. non-abusive content on social media.
- We accommodate the sociolinguistic reality of Tigrinya social media by covering posts written in both the standard Ge’ez script and Latin transliterations, ensuring that the trained models handle actual language practices.
- We demonstrate that small, specialized models with joint multi-task learning of abusiveness, topic, and sentiment tasks outperform large frontier models, establishing a strong baseline.

2 Related Work

Datasets for Abusive Language Detection. Numerous datasets have been created for the purpose of training and evaluating models for abusive language detection in English, typically sourced from online platforms such as Twitter, Reddit, YouTube, and Wikipedia [50, 57, 7, 13, 46]. Researchers have also developed datasets for languages other than English, such as the Dutch-Bully-Corpus [24] consisting of over 85,000 abusive posts, and the Arabic dataset by Mubarak et al. [37] containing 1,100 tweets and 32,000 YouTube comments. Moreover, shared tasks such as OffensEval [61, 62], GermEval 2018 [59], and HASOC 2019 [32] have provided multilingual datasets for such tasks. Pavlopoulos et al. [43] created the Greek-Gazzetta-Corpus, consisting of 1.6 million comments, and Song et al. [49] proposed a comprehensive abusiveness detection dataset with multifaceted labels from Reddit. Furthermore, Tonneau et al. [54] introduced a dataset for hate speech detection containing 35,976 tweets, comprising instances of mixed languages such as English, Nigerian Pidgin, Hausa, and Yoruba.

Approaches to Abusive Language Detection. Approaches to abusive language detection have evolved from statistical models to deep neural architectures, with transformer-based models like BERT, RoBERTa, and the GPT-family, establishing the state-of-the-art performance across multiple benchmarks [18, 12, 29, 56, 36, 2]. For low-resource languages, cross-lingual transfer from multilingual models offers significant benefits [27, 62]. Recent advances applied generative Large Language Models (LLMs) with in-context learning and data augmentation of abusive language [6, 55]. For instance, Shin et al. [48] demonstrated that GPT-generated synthetic data enhanced offensive language detection in Korean. Zhang et al. [63] proposed an approach of Bootstrapping and Distilling LLMs for toxic content detection using a novel Decision-Tree-of-Thought prompting. Jaremko et al. [25] evaluated the performance of LLMs for implicitly abusive language detection using zero-shot and few-shot learning approaches, and analyzed the models’ ability to extract relevant linguistic features.

Multi-Task Learning for Abusiveness Detection. Recent studies have demonstrated the efficacy of multi-task learning (MTL) in enhancing abusive language detection by jointly modeling related tasks. For instance, Dai et al. [10] employed a BERT-based MTL framework to simultaneously address offensive language detection, categorization, and target identification, achieving promising results. Similarly, Zhu et al. [64] integrated Prompt tuning [6, 28] with MTL to improve detection performance across multiple datasets. Mnassri et al. [35] leveraged MTL to jointly model hate speech and emotions, showing that sentiment information improves hate speech detection performance. In a related approach, Rajamanickam et al. [45] demonstrated that joint learning of toxicity and sentiment leads to more robust models than single-task approaches.

Tigrinya Language Processing. Tigrinya (ISO 639-3: *tir*) is a Semitic language of the Afro-Asiatic family that shares linguistic features with Amharic and Tigre, and uses the Ge’ez script as a writing system [21]. There is a growing interest in computational approaches to Tigrinya, such as machine translation, part-of-speech tagging, sentiment analysis, text classification through transfer learning [51, 52, 40, 26, 53, 16, 21]. More recent progress on question answering and named entity recognition [22, 5] has been enabled by datasets and pre-trained language models [19, 20, 39].

Resources for African Languages. In a related effort, Ayele et al. [4] developed a dataset for Amharic, which contains 8,258 tweets annotated for hate/offensive category, target type, and intensity on a continuous scale. Contemporary to our work, Muhammad et al. [38] introduced AfriHate, a large-scale Twitter-based benchmark dataset for hate and abusive language in 15 African languages, including a Tigrinya (*tir*) subset with 5,072 tweets annotated for abusiveness and target types. The Tigrinya slice shows a class imbalance typical of keyword-retrieval pipelines and a moderate inter-annotator agreement. Baseline experiments on AfriHate demonstrate that multilingual fine-tuned models reach a macro F1 of 74.5%, outperforming few-shot prompting of GPT-4o [41] by 18 points. In TiALD, we employ a semi-automated data-driven sampling strategy designed to diversify the annotation pool, resulting in a more challenging and representative benchmark. We sourced data from YouTube due to its substantially higher popularity among Tigrinya speakers in Eritrea and Ethiopia compared to Twitter (X) and other social media platforms.³ Furthermore, YouTube comments provide rich context, including video titles and descriptions, beneficial for modeling abusive content detection. Thus, TiALD and AfriHate are complementary resources for the research community with different characteristics. To the best of our knowledge, there is no existing resource that simultaneously addresses the abusiveness, sentiment, and topic tasks in a low-resource setting. Our work addresses this gap by providing a vital resource for content moderation practitioners, but also a benchmark for the broader research community.

3 Dataset Construction

In this section, we describe the construction of the TiALD dataset and an analysis of the annotations.

3.1 Data Collection and Preprocessing

As the source of data, we initially collected 4.1 million comments from 52 popular YouTube channels with more than 34.5K videos and a total of over 2.2 billion views at the time of collection. These channels cover various genres including news, entertainment, music, educational, documentaries, vlogs, and more, ensuring a diverse and representative sample of the Tigrinya-speaking social media landscape. We then preprocessed the data by filtering out non-text comments such as those that contain only emojis and also non-Tigrinya comments, such as those written fully in English, Arabic, Amharic, etc, using the GeezSwitch library [21]. Within the Tigrinya content, we observed that around 64% of the collected source comments contained Romanized text, where users employ improvised and non-standard transliteration schemes. To accommodate this, we cover comments written in both scripts in the annotation process, 70% Ge’ez and 30% Latin or mixed, which could help develop models that reflect a realistic usage of the language in social media.

3.2 Data Samples Selection

Naturally, abusive content constitutes a minority proportion compared to non-abusive or benign content, which implies that random sampling would be biased towards the majority content type. To avoid this sampling bias, typically a fixed list of abusive or toxic words is used in the literature to search for texts with those terms. However, this approach has a limitation as it leads to the lack of diversity in the final samples, especially for languages where there is no extensive curation of such words. Recognizing the need for a method that suits low-resource settings, we employed a semi-automatic strategy that leverages a vector space of candidate samples and a small set of terms to effectively expand and the selection criteria without heavily relying on lexical search of a fixed list of

³Market shares of YouTube vs. Twitter (X) between Jan 2018 and Mar 2025 in Eritrea (36.36% vs. 9.09%) and Ethiopia (13.08% vs. 6.81%): <https://gs.statcounter.com/social-media-stats/all/eritrea/#monthly-201801-202503>, <https://gs.statcounter.com/social-media-stats/all/ethiopia/#monthly-201801-202503>.

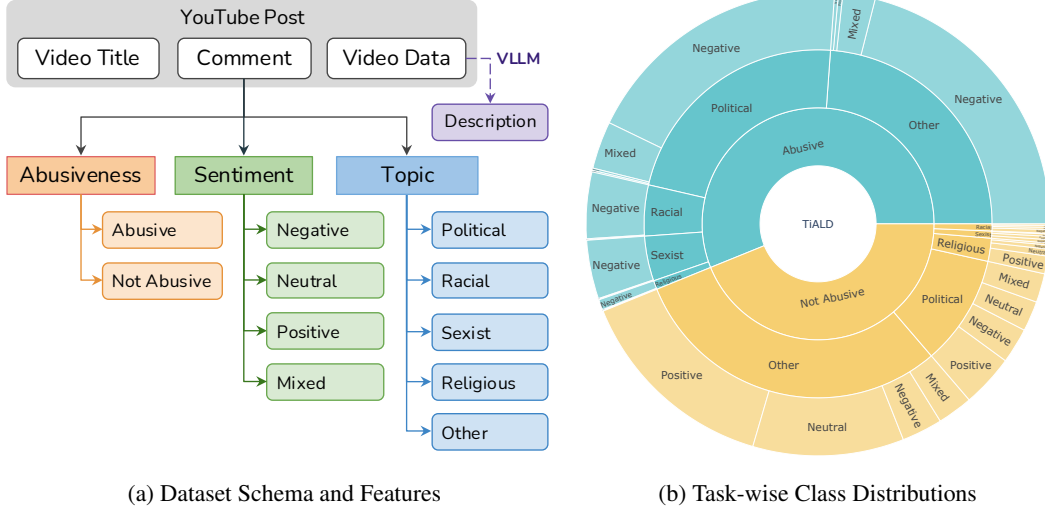


Figure 1: An overview of the TiALD Dataset: Annotation Schema and Distributions.

hand curated terms for selection. To this end, we trained word embeddings on the original 4.1 million comments we collected from YouTube, using word2vec [34] and vector dimension of 300. Then we applied an iterative process of term clustering and de-duplication to select viable comments for the annotation.

Seed Word Selection and Iterative Expansion. As a first step, we manually selected seed words for the target classes: abusive, non-abusive, political, religious, etc. These seed words included not only derogatory terms but also common words, political party names, races, and religious terms among others. Using the seed words, we expanded the list by iteratively computing their nearest k neighbors in the word embedding vector space in three stages, i.e., subsequently searching for 50, 25, and 10 semantically similar but morphologically different words for every term in each stage, respectively. This process led to an expanded list of 8,728 diverse and representative words. We then selected 15K comments covering at least two instances for every word in the expanded list, and further added 5K randomly sampled comments from the remaining data as a control group. The result is a balanced annotation pool of 20K comments with high lexical diversity (type-to-token ratio of 0.28 vs. 0.13 for random sampling) that was then made available for the annotation process.

3.3 Data Annotation

We hired nine native speakers as annotators, four females and five males, between the ages of 22 to 48 years. The annotators were asked to label each comment for three tasks: abusiveness, sentiment, and topic. For **Abusiveness**, comments are categorized into two classes (abusive or not abusive), providing the primary classification target. The **Sentiment** dimension adds emotional context with four possible classifications (positive, neutral, negative, or mixed). Finally, the **Topic** classification assigns each comment to one of five categories (political, racial, sexist, religious, or miscellaneous topics), capturing the subject matter of the context. The annotation schema of the three tasks is depicted in Figure 1a. The annotation campaign was conducted in a controlled setting with informed consent from each participant and a set of instructions and annotation guidelines were provided to ensure the quality and consistency of the dataset. By the end of the process, we collected annotations for 13,717 comments and we measure the inter-annotator agreement as discussed in Section 3.4.

Generating Video Descriptions. To provide richer contextual information for analysis, we extended the dataset with descriptions of the visual content in the videos corresponding to comments in the evaluation splits. These descriptions enable researchers to investigate potential relationships between video content and abusive language, such as whether certain visual elements might trigger hostile comments. We generated these descriptions using the Qwen-2.5-VL 3B [44] and refined them with GPT-4o [41] to ensure quality and consistency, creating a unique multimodal dimension for studying

Table 1: TiALD Dataset: Distribution of the Three Tasks and Dataset Splits.

Task	Label	Train	Test	Dev	Samples
Abusiveness	Abusive	6,980	450	250	7,680
	Not Abusive	5,337	450	250	6,037
Sentiment	Positive	2,433	226	108	2,767
	Neutral	1,671	129	71	1,871
	Negative	6,907	474	252	7,633
	Mixed	1,306	71	69	1,446
Topic	Political	4,037	279	159	4,475
	Racial	633	113	23	769
	Sexist	564	78	21	663
	Religious	244	157	11	412
	Others	6,839	273	286	7,398
	Total	12,317	900	500	13,717

contextual factors in online abuse detection. See Appendix D for the model instructions used and other details about in this step.

3.4 Inter-Annotator Agreement (IAA)

To compute IAA scores, we randomly sampled 100 comments from each annotator’s contributions (a total of 900 samples) and then asked each of the nine annotators to provide secondary labels to comments that were not initially annotated by them. Then each comment in the sample was rated by two different annotators, resulting in a total of 1,800 annotations for each of the three tasks. This approach enabled us to compute the agreement between the pairs of contributors using Cohen’s Kappa [8]. The aggregate scores for each task are: $\kappa = 0.758$ for *Abusiveness*, $\kappa = 0.649$ for *Sentiment*, and $\kappa = 0.603$ for *Topic* annotations. According to Cohen’s interpretation, these scores indicate a *substantial agreement* for abusiveness detection and sentiment analysis annotations (i.e., $0.61 \leq \kappa \leq 0.80$) and a *moderate agreement* for topic classification (i.e., $0.41 \leq \kappa \leq 0.60$). We assessed 25 random comments that were assigned different Topic labels by the annotators and found that most disagreements were due to the potential applicability of the comments to multiple topics.

3.5 Gold-label Adjudication for Evaluation

To construct a high quality test set, we extended the double-annotation process to three annotators and determined a gold label for each of the 900 samples. Our analysis showed that the initial two annotators achieved a full agreement on 546 comments, while they disagreed in at least one task for the remaining 354 samples. To adjudicate the differences, we hired additional two experts who reviewed the inconsistent annotations of the initial annotators and decided on the final labels, which are considered as the gold labels in our test set.

Finally, we partitioned the remaining single-annotated samples into two splits for training and validation (500 samples), maintaining stratified proportions of the classes for abusiveness, sentiment, and topic in each split. Table 1 presents the detailed sample distribution across the splits and Figure 1b depicts the overall class distribution of the TiALD dataset for the three tasks.

4 Experimental Setup of Baselines

To establish strong baselines for abusive language detection in Tigrinya, we evaluate three complementary approaches. First, we conduct single-task fine-tuning by training and evaluating several pre-trained language models (PLMs) on each classification task individually. Second, we implement a multi-task joint learning framework, where a shared encoder model simultaneously learns all three tasks through task-specific output heads. Finally, we assess the zero- and few-shot capabilities of state-of-the-art generative LLMs on the abusiveness detection task using prompts.

4.1 Single-task Fine-tuned Baseline Models

For each task in TiALD, we fine-tune several monolingual and multilingual PLMs that offer varying levels of adaptation to Tigrinya and other African languages. These include: monolingual models **TiRoBERTa** (125M) and **TiELECTRA** (14M) [20] trained exclusively on Tigrinya texts; multilingual models **AfriBERTa-base** (112M) [39] and **AfroXLMR-Large-76L** (560M) [1], pre-trained on 11 and 76 African languages, respectively; and **XLM-RoBERTa-base** (279M) [9], a general-purpose multilingual model trained on 100 languages as a control.

4.2 Joint Multi-task Learning Setup

In our joint learning setup, we employ a single transformer encoder shared across the three tasks that simultaneously learns to categorize the input content according to all relevant labels. Formally, let $\mathbf{h} = \text{Encoder}(x) \in \mathbb{R}^d$ denote the contextualized representation of an input comment x , given by the final hidden state corresponding to the model’s classification token (e.g., ‘[CLS]’ for BERT or ‘<s>’ for RoBERTa). A single linear classification head then maps $\mathbf{h}$ to a vector of logits:

$$\mathbf{z} = W \mathbf{h} + \mathbf{b}, \quad W \in \mathbb{R}^{L \times d}, \quad \mathbf{b} \in \mathbb{R}^L,$$

where $L = 2 + 4 + 5 = 11$ is the total number of labels covering (i) abusiveness (binary), (ii) sentiment (4-way), and (iii) topic (5-way) tasks in the TiALD dataset. Each logit z_j is passed through a sigmoid to produce the probability of label j ; a threshold of 0.5 is applied during inference to obtain binary predictions:

$$\hat{y}_j = \sigma(z_j), \quad j = 1, \dots, L.$$

Training minimizes the average binary cross-entropy (BCE) loss over all label-example pairs:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^L \left[y_{ij} \log \hat{y}_{ij} + (1 - y_{ij}) \log (1 - \hat{y}_{ij}) \right],$$

where N is the total number of examples in the training set and $y_{ij} \in \{0, 1\}$ indicates the presence of label j in the i th example.

This hard-parameter-sharing approach treats each label as an independent binary predictor while leveraging a shared representation across all three tasks, encouraging the model to capture features that benefit abusive language detection, topic classification, and sentiment analysis simultaneously. In our baseline setup, all labels contribute equally to the loss; future work may explore per-task or per-class weighting to address label imbalance.

4.3 Prompt-based Evaluation of LLMs

To assess the capabilities of state-of-the-art generative LLMs on Tigrinya abusive language detection, we employ prompt-based zero- and few-shot in-context learning [6, 58]. We design prompt templates that include either no examples (zero-shot) or a small set of annotated examples (few-shot) randomly sampled from the training set. We evaluate two commercial frontier models, **GPT-4o** [41] and **Claude Sonnet 3.7** [3],⁴ and two smaller open-weight models, **LLaMA-3.2 3B** [33] and **Gemma-3 4B** [23], for comparison. To account for variability, we run the predictions twice and compute the average scores. All input comments are preserved in their original script (Ge’ez, Latin, or mixed), and the prompt explicitly mentions that the comment is in Tigrinya. See Appendix C for the full template of the instructions used in our experiments.

Evaluation metrics. We report model performance using *Macro F1 Score* at the task level, giving equal weight to all classes; *Per-Class F1 Score* to assess the effect of class imbalance across tasks; and *Accuracy* as a secondary metric to capture overall performance at the task level.

Model training settings. For single-task and multi-task fine-tuning experiments, we set the maximum input length to 256 tokens when using comment text only and 384 tokens when incorporating video titles. We use a learning rate of $2e^{-5}$, a batch size of 16, and train for a maximum of six epochs with early stopping based on validation macro F1 score (patience of 3). We employ the

⁴OpenAI’s GPT-4o (gpt-4o-2024-08-06) and Anthropic’s Sonnet 3.7 (claude-3-7-sonnet-20250219).

Table 2: Performance in Single and Multi-task settings. Models are trained and evaluated with comment text only. The *TiALD Score* is the average of the macro F1 scores across the three tasks. The task-level highest scores for each approach are underlined, and the overall best scores are in **bold**.

Model	Abusiveness		Sentiment		Topic		TiALD Score
	Acc	F1	Acc	F1	Acc	F1	Macro F1
Single-task Models							
TiELECTRA-small	82.33	82.33	66.56	42.39	51.44	26.90	50.54
TiRoBERTa-base	<u>86.67</u>	<u>86.67</u>	66.67	52.82	<u>62.00</u>	<u>54.23</u>	<u>64.57</u>
AfriBERTa-base	83.44	83.42	66.00	50.81	61.22	53.20	62.48
Afro-XLMR-Large-76L	85.22	85.20	<u>68.56</u>	<u>54.94</u>	61.00	51.42	63.86
XLM-RoBERTa-base	81.11	81.08	<u>59.33</u>	<u>30.17</u>	58.44	43.97	51.74
Multi-task Joint Learning							
TiELECTRA-small	84.22	84.21	67.33	43.44	51.00	29.27	52.30
TiRoBERTa-base	<u>86.11</u>	<u>86.11</u>	65.33	53.41	<u>61.56</u>	<u>54.91</u>	<u>64.81</u>
AfriBERTa-base	83.67	83.66	66.89	50.19	61.55	53.49	62.45
Afro-XLMR-Large-76L	85.44	85.44	<u>68.56</u>	<u>54.50</u>	60.67	52.46	64.13
XLM-RoBERTa-base	79.89	79.87	66.67	45.40	54.22	35.50	53.59

AdamW optimizer [31] and implement our training system with PyTorch [42] and the Hugging Face Transformers library [60]. This experimental setup provides comprehensive benchmarks for evaluating single-task, multi-task, and generative approaches to abusive language detection in a low-resource setting.

5 Results and Analysis

5.1 Performance of Fine-tuned Monolingual and Multilingual Models

Our experimental results demonstrate that jointly training models on all three tasks consistently enhances performance over the single-task approaches. This improvement suggests that abusiveness, sentiment, and topic share complementary linguistic features that benefit from unified representation learning. As shown in Table 2, Tigrinya-specific models outperform general multilingual alternatives. TiRoBERTa-base achieves the highest macro F1 scores across most settings (86.67% for abusiveness detection, 54.23% for topic classification), demonstrating the value of language-specific pre-training. The Africa-centric AfroXLMR-76L model performs competitively, particularly for sentiment analysis where it reaches the highest F1 score of 54.94%, suggesting that well-adapted multilingual models can reach monolingual performance.

Multi-task joint learning improves performance across almost all models and tasks, with the most substantial gains observed for TiELECTRA-small (+1.76 percentage points in overall TiALD score) and XLM-RoBERTa-base (+1.85 points). The consistent improvement across diverse model architectures confirms that the complementary signals in the TiALD tasks can be leveraged through parameter sharing. Notably, XLM-RoBERTa-base consistently underperformed compared to both the Tigrinya-specific and Africa-adapted models, with a substantial 12 percentage point average gap on the aggregate macro F1 scores. This performance disparity highlights the limitations of general multilingual models when applied to low-resource languages with unique linguistic characteristics.

5.2 Performance of Large Language Models

The results in Table 3 reveal that even state-of-the-art LLMs struggle to match the performance of small fine-tuned models on Tigrinya abusive language detection. GPT-4o achieves 71.05% F1 with zero-shot prompting, which, while impressive, still falls 15 percentage points behind the tuned TiRoBERTa-base. The performance gap between zero-shot and few-shot settings varies dramatically across models. Sonnet 3.7 shows a large improvement (59.20% $\rightarrow$ 79.31%), suggesting high sensitivity to in-context examples. By contrast, the smaller open-weight models show degradation in the few-shot setting. In particular, LLaMA-3.2 3B struggles significantly with Tigrinya texts. It classified 96.9% of the queries as *abusive* in the zero-shot setting and produced invalid responses

Table 3: Performance of LLMs on Abusiveness detection task with and without contextual information, video_title and video_description. †LLaMA-3.2 3B produced invalid responses for over 61% of queries in both few-shot settings, mainly due to its limited Tigrinya text understanding.

Model	Comment Only				Video Context + Comment			
	Zero-shot		Few-shot		Zero-shot		Few-shot	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Closed Frontier Models								
GPT-4o	71.50	71.05	72.17	72.06	74.72	74.70	74.67	74.53
Claude Sonnet 3.7	63.89	59.20	79.33	79.31	73.00	72.02	78.56	78.21
Open-weight Models								
Gemma-3 4B	61.72	61.13	54.78	49.33	56.83	53.45	54.61	47.13
LLaMA-3.2 3B†	50.83	36.93	19.28	18.91	50.28	48.26	19.06	18.79

for over 61% of the queries in the few-shot setting. These results highlight the limitations of current LLMs when processing low-resource languages outside their primary training distribution.

Leveraging Video Context for Abusiveness Detection. When enriched with contextual information (video titles and the generated video descriptions), the performance of the frontier LLMs improves notably. GPT-4o performs the best in the zero-shot settings, gaining 3.65 percentage points. Similarly, Sonnet 3.7 shows substantial improvement in zero-shot performance (+12.82 points) when provided with video context. These gains underscore the importance of contextual understanding in accurately identifying abusive content, particularly when the language itself presents challenges for the models. The consistent improvement across settings confirms our hypothesis that supplementary video context provides valuable signals for content moderation in low-resource languages.

For a more detailed breakdown of model performance, see Appendix Tables 5 and 6, which present per-class F1 scores across all experimental settings.

5.3 Analysis and Insights

Cross-Task Performance Analysis. Performance analysis reveals that models achieve higher F1 scores for detecting abusive content (79-86%) compared to sentiment analysis (30-54%) and topic classification (26-54%). This pattern likely reflects the multi-class nature of sentiment and topic annotations increasing the difficulty of the tasks, as evidenced by the lower inter-annotator agreement scores. The best-performing model, TiRoBERTa-base with joint learning setup, demonstrates a relatively balanced increase across the sentiment (+0.6) and topic (+0.7) tasks, but both remain challenging with only 53.41% and 54.91% macro F1 scores, respectively. This performance gap presents an opportunity for future research to develop more specialized approaches to sentiment and topic understanding in morphologically complex languages like Tigrinya.

Impact of Contextual Information in Fine-tuned Models. Social media comments often reference the content they respond to, making contextual information potentially valuable for understanding and classifying them. When incorporating video titles as additional context (Table 4), performance is further improved for most models. TiRoBERTa-base with video context achieves the highest overall performance (65.23% TiALD score), with significant gain (+1.5) observed primarily in sentiment classification. This suggests that the video context provides topical cues that help disambiguate the intent and subject of potentially abusive comments.

Effectiveness of Iterative Seed-Expansion Sampling. Compared to conventional fixed-vocabulary methods, our iterative seed-expansion sampling approach generates a more diverse and representative annotation pool while preserving lexical diversity, which is a crucial factor for languages with highly inflectional morphology where words can take numerous surface forms. Quantitative analysis reveals that our approach yielded higher type-to-token ratio (27.6%) compared to keyword-based sampling using existing toxic word lists (18.2%) and the source corpus (7.2%) of all the 4.1M comments. Furthermore, we analyzed the ratio of *abusive* class annotations for the comments from our iterative sampling against those in the control group via random sampling, and we observe a significant difference, 65.2% vs. 14.3%, respectively. The random sampling is biased towards majority type and

Table 4: Performance of single-task models with video title as context. Models are trained and evaluated with comment and video title combined as input.

Model	Abusiveness		Sentiment		Topic		TiALD Score
	Acc	F1	Acc	F1	Acc	F1	Macro F1
TiELECTRA-small	81.67	81.67	65.11	39.40	53.44	27.81	49.62
TiRoBERTa-base	86.22	86.17	67.11	54.97	62.44	54.55	65.23
AfriBERTa-base	82.44	82.44	65.33	51.33	61.56	52.10	61.96
Afro-XLMR-Large-76L	84.22	84.20	67.67	52.64	62.56	54.11	63.65
SLM-RoBERTa-base	75.11	75.09	62.22	43.47	59.11	41.60	53.39

hence leads to more benign non-abusive comments, while the keyword sampling is biased towards the seed words and fails to produce a diverse pool of samples.

6 Conclusion

In this work, we introduce the **Tigrinya Abusive Language Detection (TiALD)**, the first large-scale multi-task benchmark dataset for abusive language detection in Tigrinya. TiALD comprises 13,717 user comments from YouTube videos, annotated by native speakers for three tasks: abusiveness, sentiment, and topic classification, providing a rich resource for understanding and detecting harmful content in this understudied language’s social media. Our comprehensive experiments establish strong baselines while revealing substantial room for improvement across all tasks, with performance gaps of 15-45 percentage points from perfect classification, indicating that TiALD will serve as a challenging benchmark for future research. Our analysis reveals three key insights: first, we demonstrate that small, specialized Tigrinya models substantially outperform the current frontier models in the low-resource setting; second, we show that joint multi-task learning in aggregate outperforms single-task approaches, indicating that abusiveness, sentiment, and topic share complementary signals; third, incorporating auxiliary visual content descriptions further enhances abusiveness detection performance. We make the TiALD dataset and trained models publicly available to advance content moderation research for Tigrinya-speaking communities and to serve as a blueprint for similar efforts in other low-resource languages, promoting more inclusive and effective online safety systems.

Limitations

Explicit vs. Implicit Toxicity: As the first work of its kind for Tigrinya, our work focuses on overt forms of offensive language. However, we acknowledge that implicit forms of toxicity, such as microaggressions and subtle forms of prejudice, also contribute to online harassment and should be addressed in future work. **Granular Annotation of Abusiveness:** Our dataset includes a single label for abusiveness, which may not capture the full range of abusive language. Future work could benefit from a more granular annotation scheme that captures different types of abuse, such as hate speech, harassment, and offensive language.

Ethics Statement

This research adheres to the academic and professional ethics guidelines of our institution, obtaining its Institutional Review Board (IRB) approval.⁵ All data collection and annotations were conducted with the informed consent and respect of the participants. While the development of abusive language detection systems has the potential to improve the online experience of millions of social media users worldwide, it is essential to consider the possible societal and ethical implications of such research. **Fairness and bias mitigation:** We carefully designed data collection and annotation procedures to minimize biases and avoid reinforcing stereotypes in the dataset and baseline models. **Respecting privacy:** We adhere to strict privacy guidelines while collecting and using user-generated data, ensuring that any personal information remains anonymized and protected. **Balancing moderation and expression:** We recognize the tension between detecting harmful content and protecting free expression, emphasizing that systems built on our dataset should incorporate transparent, accountable processes to minimize over-censorship. By addressing these considerations and making our dataset publicly available, we aim to contribute to safer online environments for Tigrinya speakers while providing a foundation for ethical content moderation research.

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Appendix

A Class-level Performance of Baseline Models

Table 5 presents the per-class F1 scores of the fine-tuned models across tasks. For abusiveness detection, performance is relatively balanced between *abusive* and *not abusive* classes, with TiRoBERTa-base achieving the highest scores (86.52% and 86.81% in single-task setting). In contrast, sentiment and topic classification show significant class imbalances. For sentiment, models perform strongest on the *negative* class (up to 81.32%) but struggle with *neutral* and *mixed* classes (as low as 1.53% and 0%). Similarly, *topic* classification shows strong performance on *political* content (up to 71.21%) but substantially weaker results on minority classes like *sexist* and *racial*. Notably, multi-task learning improves performance on challenging classes - for instance, TiRoBERTa-base’s F1 score on *sexist* content increases from 31.78% to 46.30% when using joint learning, highlighting the complementary nature of these tasks.

Table 5: Per-class Performance in Single and Multi-task settings in F1 score. Models are trained and evaluated with comment text only. The class-level highest scores for each approach are underlined, and the overall best scores are in **bold**.

Model	Abusiveness		Sentiment				Topic				
	Abusive	Not Abusive	Positive	Neutral	Negative	Mixed	Political	Racial	Sexist	Religious	Other
Single-task Models											
TiLECTRA-small	82.35	82.31	62.84	4.48	80.88	21.36	67.48	6.50	0.00	0.00	60.51
TiRoBERTa-base	86.52	86.81	68.68	40.00	81.18	21.43	<u>70.86</u>	46.49	31.78	<u>56.90</u>	<u>65.15</u>
AfriBERTa-base	84.00	82.85	63.25	33.78	81.03	25.17	69.64	40.46	<u>34.29</u>	56.89	64.71
Afro-XLMR-Large-76L	85.74	84.66	<u>69.44</u>	<u>41.06</u>	81.32	<u>27.94</u>	70.71	39.75	28.00	54.38	64.29
XLM-RoBERTa-base	80.32	81.84	<u>46.03</u>	1.53	73.11	0.00	67.32	33.33	2.53	52.53	64.12
Multi-task Joint Learning											
TiLECTRA-small	84.67	83.75	62.68	14.39	81.10	15.58	65.17	6.84	7.41	7.36	59.56
TiRoBERTa-base	<u>86.13</u>	<u>86.10</u>	62.98	42.75	79.83	28.07	71.21	<u>45.96</u>	46.30	47.44	63.65
AfriBERTa-base	83.93	83.39	65.39	27.43	<u>81.13</u>	26.79	70.59	44.16	44.00	43.35	65.36
Afro-XLMR-Large-76L	85.16	85.71	71.79	33.88	80.96	31.34	69.02	36.60	35.64	57.78	63.27
XLM-RoBERTa-base	80.43	79.31	67.06	15.47	79.92	19.15	67.51	16.26	16.47	15.29	61.95

Similarly, the per-class F1 scores of LLMs in Table 6 reveal distinct patterns in performance across *abusive* and *not abusive* classes. Frontier models show substantial improvements with the video’s contextual information as GPT-4o’s F1 score on *abusive* content increases from 69.04% to 76.66%. Interestingly, Sonnet 3.7 exhibits highly asymmetric zero-shot performance (44.18% on *abusive* vs. 72.58% on *not abusive* content), but dramatically improves with few-shot learning, achieving the highest overall scores for both classes (79.26% for *abusive* content in comment-only setting and 81.11% for *not abusive* with video context). Open-weight models struggle significantly with Tigrinya, particularly LLaMA-3.2 3B, which achieves only 6.36% F1 on *not abusive* content in zero-shot settings, as it classifies 96.9% of the comments as *abusive*. These results demonstrate that even current frontier models require contextual prompting to approach the performance of fine-tuned specialized models for low-resource languages.

Table 6: Per-class Performance of LLMs on the Abusiveness Detection task in F1 score. We evaluate with and without contextual information, *video_title* and *video_description*. [†]LLaMA-3.2 3B produced invalid responses for over 61% of queries in both few-shot settings, mainly due to its limited Tigrinya text understanding.

Model	Comment Only				Video Context + Comment			
	Zero-shot		Few-shot		Zero-shot		Few-shot	
	Abusive	Not Abusive	Abusive	Not Abusive	Abusive	Not Abusive	Abusive	Not Abusive
Closed Frontier Models								
GPT-4o	69.04	75.89	74.82	71.41	76.66	74.83	76.51	73.03
Claude Sonnet 3.7	44.18	72.58	79.26	80.69	64.81	76.60	75.79	81.11
Open-weight Models								
Gemma-3 4B	65.88	<u>56.27</u>	32.28	<u>66.06</u>	41.64	<u>66.37</u>	<u>28.27</u>	<u>67.10</u>
LLaMA-3.2 3B [†]	<u>66.72</u>	6.36	24.66	12.93	<u>58.47</u>	38.57	14.18	23.90

B TiALD Dataset Features

Table 7 presents the descriptions of the fields in the TiALD dataset. Figure 2 depicts the task-wise class distribution across the tasks of Abusiveness, Sentiment, and Topic.

Table 7: An overview of the features included in the TiALD Dataset.

Feature	Data Type	Description
sample_id	String	Unique identifier for the sample in the dataset.
comment_id	String	Unique identifier for the comment.
comment_original	String	Original comment text as posted by user.
comment_cleaned	String	Pre-processed version of the comment text.
abusiveness	Categorical	Abuse label (<i>Abusive</i> or <i>Not Abusive</i>).
sentiment	Categorical	Sentiment label (<i>Positive</i> , <i>Neutral</i> , <i>Negative</i> , or <i>Mixed</i>).
topic	Categorical	Topic label (<i>Political</i> , <i>Racial</i> , <i>Sexist</i> , or <i>Religious</i> , <i>Other</i>).
annotator_id	String	Identifier of the annotator.
comment_script	Categorical	Script used in the comment (<i>Ge'ez</i> , <i>Latin</i> , or <i>Mixed</i>).
comment_publish_date	String	Year and month the comment was published.
video_id	String	Identifier of the video the comment was posted under.
video_title	String	Title of the associated video.
video_num_views	Numeric	Number of views the video received.
video_publish_year	Numeric	Year the video was published.
video_description	String	Auto-generated description of the video content.
channel_id	String	Identifier of the YouTube channel the video belongs to.
channel_name	String	Name of the YouTube channel the video belongs to.

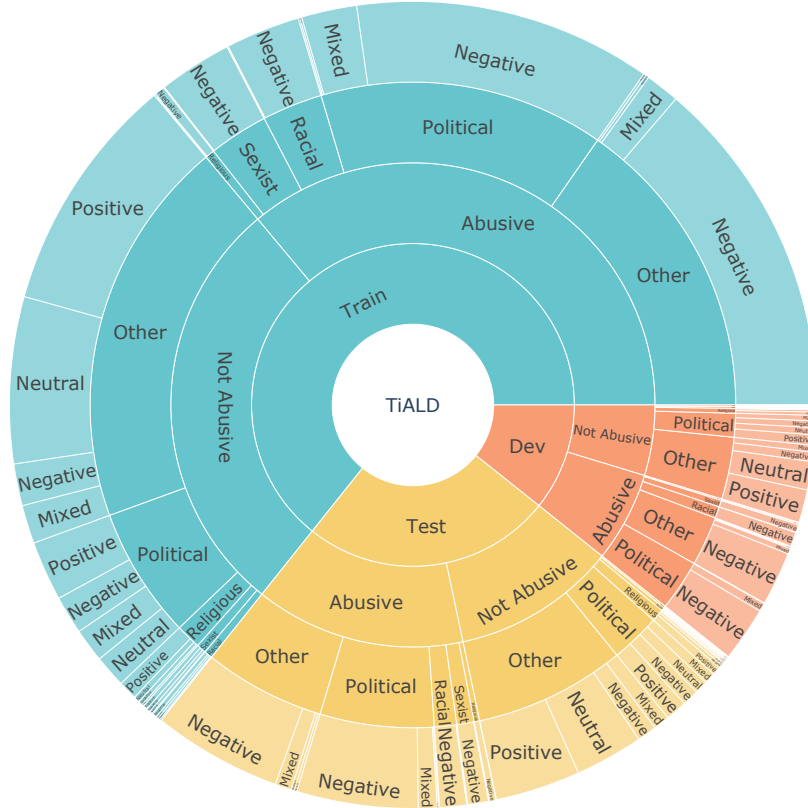


Figure 2: TiALD Class Distribution across the Splits and Tasks of Abusiveness, Sentiment, and Topic.

C Detailed LLM-Evaluation Instructions

Experiments were conducted on two commercial frontier models and two open-weight LLMs under both zero-shot and few-shot settings. All input comments were preserved in their original script (Ge’ez, Latin, or mixed), with explicit instruction that the comment was written in Tigrinya. For the few-shot evaluation, we included four balanced examples (two abusive, two non-abusive) randomly sampled from the training set and arranged in alternating order as shown below:

```
[Instruction]
Classify the following user comment written in Tigrinya as “Abusive”
or “Not Abusive”. Do not provide any explanation or additional text.
[Optional In-context Examples]
Here are some examples: <few-shot examples>
<comment text>: Abusive
<comment text>: Not Abusive
<comment text>: Abusive
<comment text>: Not Abusive
[Comment]
Comment: <comment text>
```

D Generating Video Content Descriptions

We used a vision-language model, Qwen2.5-VL-3B [44], to generate detailed descriptions of video content corresponding to comments in the evaluation splits of the TiALD dataset. Qwen2.5-VL handles long-form videos up to several hours by applying dynamic resolution processing and absolute time encoding. The TiALD dataset’s videos average 28.1 minutes (1,686 seconds) in length and can run as long as 334.75 minutes (20,085 seconds) at 30 FPS. To ensure high-quality descriptions, we trimmed videos exceeding 20 minutes to their first 20-minute segment. Each resulting clip was then passed to the model with the following instruction:

```
[Instruction]
Describe the content of this video frame in detail. Focus on
the people, objects, actions, and setting visible in the image.
Provide a comprehensive description that could help understand
what the video is about.
[Video]
<video frames>
```

To enhance the quality and consistency of the generated video descriptions, we revise them using a larger more capable model, GPT-4o [41], with the following instruction:

```
[Instruction]
Revise the following automatically generated video description
to make it clearer and consistent, while preserving the key
information. Keep your response to a moderate length up to 150
words and focus only on the video content.
[Video Title]
Video Title: <video title>
[Video Description]
Video Description: <video description>
```

Finally, the resulting video descriptions were included in the dataset as auxiliary features to enable deeper analysis of potential relationships between video content and abusiveness of comments. We also empirically show the benefit of using contextual information in our experiments, as shown by the results in Table 3.

E Annotation Guidelines for TiALD Dataset

This section outlines the detailed annotation guidelines provided to the native Tigrinya speakers who participated in the TiALD dataset creation. Annotators were instructed to classify each YouTube comment across three dimensions: Abusiveness, Sentiment, and Topic, while following specific protocols for comment eligibility.

Task 1: Abusive Language Detection

Annotators were asked to determine whether a comment contained abusive language according to the following criteria:

- **Abusive:** The comment contains language that attacks, insults, demeans, or threatens an individual or group. This includes hate speech, profanity directed at others, derogatory terms, threats of violence, severe personal attacks, or language intended to humiliate or degrade.
- **Not Abusive:** The comment does not contain language that attacks an individual or group. It may express disagreement, criticism, or negative opinions without using abusive language toward others.

Task 2: Sentiment Analysis

Annotators classified the emotional tone of each comment into one of four sentiment categories:

- **Positive:** The comment expresses primarily positive emotions, approval, praise, gratitude, happiness, or optimism. This includes congratulatory messages, expressions of joy, or positive feedback.
- **Neutral:** The comment does not express a clear positive or negative sentiment. This includes factual statements, questions without emotional content, or balanced objective comments.
- **Negative:** The comment expresses primarily negative emotions, disapproval, criticism, anger, sadness, or pessimism. This includes expressions of disappointment, frustration, or negative judgments.
- **Mixed:** The comment contains a relatively balanced mix of both positive and negative sentiments, with neither clearly dominating. This includes comments expressing contrasting emotions or evaluating different aspects both positively and negatively.

Task 3: Topic Classification

Annotators classified each comment into one of five topical categories:

- **Political:** Comments discussing political figures, parties, governments, policies, elections, or expressing political opinions. This includes references to specific political events, governance issues, or politically divisive topics.
- **Racial:** Comments referring to racial or ethnic identity, characteristics, or relationships between racial/ethnic groups. This includes discussions about cultural identity tied to ethnicity.
- **Sexist:** Comments discussing gender roles, gender identity, or containing gendered language. This includes content related to expectations based on gender or discussions about gender relations.
- **Religious:** Comments discussing religious beliefs, practices, institutions, or figures. This includes references to religious texts, doctrines, religious communities, or spirituality.
- **Other:** Comments that don't primarily fall into any of the above categories. This includes everyday conversations, entertainment discussions, personal updates, or general information sharing.

Comment Eligibility Criteria

Annotators were instructed to exclude comments from annotation if they met any of the following conditions:

1. Comments written entirely in a language other than Tigrinya (e.g., English, Amharic, Arabic)
2. Comments containing no actual Tigrinya words (e.g., consisting only of repeated characters, symbols, or emojis)
3. Comments that were unintelligible or lacked meaningful content

However, annotators were instructed to retain comments if they:

- Contained at least some Tigrinya words, even if mixed with words from other languages
- Used Romanized Tigrinya (Latin script) rather than the native Ge'ez script
- Contained code-switching between Tigrinya and other languages

These guidelines were designed to create a dataset that accurately reflects the linguistic and cultural nuances of abusive language in Tigrinya social media content.